

# Power for Agreement (Kappa)

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## Introduction

Cohen's kappa measures agreement beyond chance between two raters on a categorical scale and is the de facto standard for inter-rater reliability with categorical outcomes. Sample-size planning for a kappa-based reliability study addresses one of two distinct questions: how many subjects are needed to estimate kappa with a desired precision (a target 95 % confidence-interval half-width around the expected value), or how many subjects are needed to test a hypothesis about kappa (typically that kappa exceeds a benchmark value such as 0 or 0.6) with adequate power. Modern reliability-study protocols are increasingly precision-driven rather than hypothesis-driven, because a “kappa is greater than zero” claim conveys little practical information about a clinical instrument.

## Prerequisites

A working understanding of Cohen's kappa, the marginal-prevalence dependence of kappa's variance, and the distinction between precision-based and hypothesis-test sample-size questions.

## Theory

For an expected kappa  $\kappa_1$  with marginal prevalences  $\pi_1$  and  $\pi_2$  for the two raters, the asymptotic variance of the sample kappa is given by the Fleiss-Cicchetti-Everitt formula. Precision-based planning targets a 95 % CI half-width  $w$ :

$$n \approx \frac{(1 - \kappa_1^2) z_{0.975}^2}{w^2(1 - p_e)^2},$$

with  $p_e$  the chance-expected agreement. Hypothesis-test planning targets the power to reject  $H_0 : \kappa = \kappa_0$  in favour of  $H_1 : \kappa = \kappa_1$ , using a Normal-approximation  $z$ -test on the difference between estimated and hypothesised values, scaled by the appropriate standard errors.

## Assumptions

Subjects are independent and each is rated once by each of the two raters; the expected kappa and the marginal prevalences are pre-specified from pilot data or substantive expectations; the rating scale is categorical (or ordinal with weighted kappa).

## R Implementation

```
library(irr)

N.cohen.kappa(rate1 = 0.5, rate2 = 0.5,
              k1 = 0.70, k0 = 0,
              alpha = 0.05, power = 0.80,
              twosided = TRUE)

kappa_exp <- 0.70; p_e <- 0.5
```

```
w <- 0.05
n_prec <- qnorm(0.975)^2 * (1 - kappa_exp^2) / (w^2 * (1 - p_e)^2)
ceiling(n_prec)
```

## Output & Results

`N.cohen.kappa()` returns the sample size for the hypothesis test of  $\kappa_1$  vs  $\kappa_0$  given the marginal rates. For an expected  $\kappa_1 = 0.70$  against  $\kappa_0 = 0$  at balanced marginals, roughly 40 subjects suffice for the hypothesis-test approach; precision-targeted with a 95 % CI half-width of 0.05 requires roughly 83 subjects, illustrating that precision-based planning typically demands more subjects than hypothesis-test planning.

## Interpretation

A reporting sentence: “To estimate inter-rater Cohen’s kappa with a 95 % CI half-width of 0.05 around an anticipated value of 0.70 (assumed marginal prevalences 0.5 in each rater), 83 subjects are required. The protocol enrolls 100 to allow for 15 % unevaluable ratings. A hypothesis-test framing — power to reject  $\kappa = 0$  at 80 % power — would require only 40 subjects, but the precision-based plan is preferred because the substantive question is the magnitude of agreement, not whether it differs from zero.” Always state the planning paradigm.

## Practical Tips

- Kappa’s variance depends substantially on the marginal prevalences; very imbalanced marginals (e.g., one category at 5 % prevalence) inflate the variance and require more subjects than balanced marginals at the same expected kappa value.
- Weighted kappa for ordinal categories has smaller standard error than unweighted kappa for nominal categories with the same number of categories; adjust the sample-size calculation accordingly using the weighted-kappa variance formula.
- Multi-rater kappa (Fleiss’s kappa) requires a different sample-size calculation than two-rater Cohen’s kappa; use `samplesize::kappa.sample.size()` or simulation-based approaches for the multi-rater design.
- Always report the marginal prevalences and the assumed agreement matrix in the protocol, not just the expected kappa estimate; reviewers cannot reproduce the calculation without them, and the marginals materially affect the variance.
- For continuous ratings, the appropriate reliability statistic is the intraclass correlation coefficient (ICC), and a separate sample-size calculation applies; do not dichotomise continuous ratings to compute kappa.
- For the kappa-paradox situation where high observed agreement coincides with low kappa due to imbalanced marginals, consider reporting both kappa and the prevalence- and bias-adjusted kappa (PABAK) for a fuller picture of reliability.

## R Packages Used

`irr::N.cohen.kappa()` for the canonical Cohen’s-kappa hypothesis-test sample-size calculation; `samplesize::kappa.sample.size()` for an alternative interface and multi-rater extensions; `kappaSize` for kappa sample-size with multi-category outcomes; `simr` for simulation-based extensions; `psych` and `irr` for kappa analysis after data collection.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.

- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation